

Results - Machine Learning and System Thinking

Dec 1-31: Performance (daily averages)

Depending on the course, we can choose to maximize purchases or clicks.

Course	Strategy	Metrics					Improvement (vs Actual)		
		Clicks	Purch	Cost	Clicks/\$	Kws	Clicks	Purch	Clicks/\$
ML	Actual	146.5	0.12	332.28	0.441	38	—	—	—
	Max Purch	122.6	0.15	353.94	0.346	410	-16.3%	+30.2%	-21.4%
	Max Clicks	281.7	0.06	353.88	0.796	285	+92.2%	-5.4%	+80.5%
Sys Think	Actual	122.3	0.26	331.81	0.369	35	—	—	—
	Max Purch	265.2	0.34	357.05	0.743	119	+116.9%	+31.7%	+101.5%
	Max Clicks	252.9	0.02	356.98	0.709	107	+106.8%	-54.2%	+92.3%

Experimental Results

Dec 1-31: Regional Breakdown

Spend in each region adapts according to the course and strategy.

Course	Strategy	Spend Percentage		
		USA	Region A	Region B
ML	Actual	75.3%	15.1%	9.6%
	Max Purch	93.2%	6.8%	0.0%
	Max Clicks	33.4%	33.3%	33.2%
Sys Think	Actual	65.7%	24.3%	10.1%
	Max Purch	35.9%	35.2%	28.9%
	Max Clicks	33.6%	33.4%	33.0%

Experimental Results

Dec 1-31: Match Type Breakdown

Similarly, spend on each match type adapts according to the course and strategy. For both courses, we observe a shift towards more Exact and Phrase matches.

Course	Strategy	Spend Percentage		
		Exact	Phrase	Broad
ML	Actual	2.6%	5.2%	92.2%
	Max Purch	15.9%	23.0%	61.1%
	Max Clicks	12.9%	20.0%	67.1%
Sys Think	Actual	11.5%	7.9%	80.6%
	Max Purch	12.9%	27.7%	59.5%
	Max Clicks	17.8%	27.9%	54.3%

Implementation

Input data

- Fixed list of keywords (from Google Ads past searches, Semrush).
- Google Ads reports
 - Keyword-level past performance (daily clicks, costs).
 - Campaign-level past performance (total clicks, purchases, also broken down by segment).
 - Historical metrics for each keyword (e.g., search volume).
 - These are currently manually downloaded, but we would need to automate this to download daily (e.g., using Google Ads API).

Implementation

Processing

The processing pipeline consists of the following stages (currently four separate Python scripts):

- ➊ Data processing: combining reports, getting keyword embeddings.
- ➋ Modeling: fit an XGBoost model to predict clicks.
- ➌ Optimization: embed model and use Gurobi to solve (maximizing clicks or purchases). The model is large and is currently solved on the Engaging compute cluster.
- ➍ Post-processing: daily budgets, manual bids, and bid adjustments.

Processing — in more detail

Data processing

- ① Load daily keyword performance (clicks, cost) from Google Ads.
- ② Extract campaign → region mapping (USA, A, B) from campaign names.
- ③ Merge with Google Keyword Planner data: search volume, competition index, bid ranges.
- ④ Engineer temporal features: day-of-week, weekend, holiday, month, days to next course start.
- ⑤ Compute search-trend features: 3-month & 6-month averages, month-over-month change, trend slope.
- ⑥ Generate 50-d keyword embeddings via BERT + TruncatedSVD.
- ⑦ Impute missing values at multiple stages:
 - On load: fill missing clicks/cost with 0; drop rows with unparseable dates.
 - Before GKP merge: fill numeric nulls with 0, categorical with “Missing”.
 - After GKP merge: fuzzy-match keywords to fill remaining GKP gaps (search volume, trends).
 - Final cleanup: drop any remaining rows with NaNs.
- ⑧ Remove outliers, train/test split.

Processing — in more detail

Modeling

- Target: daily clicks per (keyword, match type, region) combination.
- XGBoost regressor (MSE loss) with StandardScaler + OneHotEncoder preprocessing.
- Hyperparameters tuned via 5-fold cross-validated grid search over:
 - `n_estimators` $\in \{5, 10, 20\}$, `max_depth` $\in \{2, 3, 4\}$, `learning_rate` $\in \{0.1, 0.3\}$.
- Features include keyword embeddings, search stats, temporal, competitiveness, match type, region, and **cost** (the decision variable).
- Model interpretability: SHAP values confirm cost, search volume, and days-to-start are top drivers.

Processing — in more detail

Optimization — maximize clicks

Decision variables.

- $x_i \geq 0$: daily cost (spend) for combination i .
- $\hat{g}_i$: predicted clicks from the embedded XGBoost model for combination i .
- $B_r \geq 0$: regional sub-budget for region r .

Formulation:

$$\begin{aligned} \max_{x, \hat{g}, B_r} \quad & \sum_i \hat{g}_i \\ \text{s.t.} \quad & \hat{g}_i = f_{\text{XGB}}(x_i, \mathbf{w}_i) && \forall i \\ & \sum_{r \in \mathcal{R}} B_r = B && \text{(total budget)} \\ & \sum_{i \in \mathcal{I}_r} x_i \leq B_r && \forall r \in \mathcal{R} \\ & \hat{g}_i - \hat{g}_i^{(0)} \leq S_i && \forall i \text{ with exact match} \\ & B_{\text{USA}} \geq B_{\text{A}} \geq B_{\text{B}} && \text{(optional)} \\ & x_i \geq 0, \hat{g}_i \geq 0 && \forall i \end{aligned}$$

where $\mathbf{w}_i$ are the static features, B is the total daily budget, B_r is the regional sub-budget, $\mathcal{I}_r$ is the set of combinations in region r , $\hat{g}_i^{(0)}$ is predicted clicks at cost 0, and S_i is the historical monthly search volume.

Processing — in more detail

Optimization — maximize purchases

If maximizing expected purchases instead of clicks, the objective changes to:

$$\max_{x, \hat{g}, B_r} \sum_{r \in \mathcal{R}} \sum_{i \in \mathcal{I}_r} \hat{g}_i \cdot p_r$$

where $p_r = \frac{\text{total purchases in region } r}{\text{total clicks in region } r}$ is the historical purchase rate for region r .

All other constraints remain the same.

XGBoost embedding. Each tree is linearized: for each row i and tree t , binary indicator variables $z_{i,t,\ell}$ select exactly one leaf. Static feature splits are pruned at build time; the cost variable splits become Big-M constraints on x_i , giving a mixed-integer program solvable by Gurobi.

Implementation

Uploading solution to Google Ads

The following should be uploaded daily to Google Ads (e.g., through API):

- ❶ Separate each match type and region into campaigns, and have a daily budget for each campaign. E.g., USA + Exact match, USA + Phrase match, etc.
- ❷ Base bid for each i (keyword, match type, region combination).
 - Calculated as $\frac{\text{daily cost}_i}{\text{predicted clicks}_i} \times 1.3$
 - $\times 1.3$ is because we pay less than what we bid due to the second-price auction mechanism. This value will need to be validated experimentally.
- ❸ Bid adjustments for hour of the day and the user.
 - Based on purchase rates $\left(\frac{\text{num purchases}}{\text{num clicks}} \right)$.
 - Calculated as $\frac{\text{segment purch rate}}{\text{region purch rate}} - 1$.
 - Input to Google Ads as percentage adjustments to the base bid (e.g., +30%, -50%).

Example input to Google Ads

ML, Dec 1st, Max Purchases: Daily Campaign Budgets

Region	Match Type	Daily Budget (\$)
USA	Broad match	179.64
USA	Exact match	75.84
USA	Phrase match	98.38
Total		353.86

Table: Daily budget by region and match type.

Example input to Google Ads

ML, Dec 1st, Max Purchases: Sorted by Daily Budget (Top 20)

Keyword	Region	Match Type	Base Bid (\$)
free ai and machine learning courses with certificates	USA	Broad match	1.81
ai training courses	USA	Broad match	1.81
ai free courses	USA	Broad match	1.81
nvidia training courses	USA	Broad match	1.81
course cs50	USA	Broad match	1.72
aws certified machine learning engineer associate course	USA	Broad match	1.61
training data machine learning	USA	Phrase match	1.83
ict iitr	USA	Phrase match	1.83
sas machine learning course	USA	Phrase match	1.83
kaggle machine learning course	USA	Phrase match	1.83
download python for data science and machine learning bootcamp course	USA	Phrase match	1.83
machine learning course certificate free	USA	Phrase match	1.83
google free machine learning course with certificate	USA	Phrase match	1.83
learn how to use a sewing machine course	USA	Phrase match	1.83
machine learning engineer course udemy	USA	Phrase match	1.83
machine learning systems design course	USA	Phrase match	1.83
machine learning system design course	USA	Phrase match	1.83
octave machine learning	USA	Exact match	1.80
scientific computing with python freecodecamp	USA	Exact match	1.80
analytics vidhya machine learning free course	USA	Exact match	1.80

Table: Example optimized bids (top 20, sorted by daily budget then base bid).

Example input to Google Ads

ML, Bid Adjustments (with clicks and purchases for reference)

Region	Segment	Value	Clicks	Purchases	Bid Adjustment
Increases					
USA	Hour of Day	15 - 17	5,942	10.5	+78%
USA	Hour of Day	12 - 14	6,554	8.7	+33%
A	Device	Computers	15,593	10.1	+31%
USA	Device	Computers	25,877	33.1	+28%
USA	Age	35 - 44	5,374	6.5	+22%
A	Age	Unknown	5,794	3.0	+5%
Decreases					
USA	Device	Mobile phones	9,426	2.5	-74%
A	Device	Mobile phones	8,615	2.0	-53%
USA	Age	18 - 24	5,633	3.0	-46%
USA	Age	25 - 34	9,078	6.0	-34%
A	Age	25 - 34	7,964	3.1	-21%
USA	Age	Unknown	9,066	8.0	-11%

Note: total of 22 non-zero bid adjustments.

Bid adjustments are only calculated where we have more than 5,000 clicks.

Example input to Google Ads

Sys Think, Dec 1st, Max Purchases: Daily Campaign Budgets

Region	Match Type	Daily Budget (\$)
A	Broad match	45.51
A	Exact match	18.46
A	Phrase match	54.77
B	Broad match	68.44
B	Exact match	21.16
B	Phrase match	29.29
USA	Broad match	93.61
USA	Exact match	12.95
USA	Phrase match	12.80
Total		356.99

Table: Daily budget by region and match type.

Example input to Google Ads

Sys Think, Dec 1st, Max Purchases: Sorted by Daily Budget (Top 20)

Keyword	Region	Match Type	Base Bid (\$)
software architect certification	USA	Broad match	3.77
microsoft project management certification	USA	Broad match	3.75
design patterns certification	B	Broad match	0.49
system design certification	B	Broad match	0.53
system design certification	B	Exact match	0.52
certificate in design thinking	B	Broad match	0.56
lms learning program	USA	Broad match	3.65
certificate in design thinking	B	Phrase match	0.54
cm2 certification	A	Phrase match	1.70
system design certification	B	Phrase match	0.63
asana academy certification	A	Phrase match	1.64
cpim certification	A	Phrase match	1.61
design patterns certification	A	Phrase match	1.59
system design certification	A	Broad match	1.50
metadata certification	A	Exact match	1.70
metadata certification	A	Phrase match	1.66
asana academy certification	A	Broad match	1.63
alison course certificate	A	Broad match	1.63
cpim certification	A	Broad match	1.59
design patterns certification	A	Broad match	1.58

Table: Example optimized bids (top 20, sorted by daily budget then base bid).

Example input to Google Ads

Sys Think, Bid Adjustments (with clicks and purchases for reference)

Region	Segment	Value	Clicks	Purchases	Bid Adjustment
Top 10 Increases					
B	Location	Brazil	12,657	7.0	+134%
A	Location	United Kingdom	7,541	14.8	+132%
B	Location	Peru	5,546	2.5	+87%
B	Age	35 - 44	20,896	13.9	+72%
B	Device	Computers	46,734	28.8	+59%
A	Device	Computers	39,772	97.5	+59%
B	Hour of Day	6 - 8	15,858	8.8	+44%
USA	Hour of Day	12 - 14	14,815	33.9	+42%
A	Age	Unknown	17,403	38.0	+41%
B	Hour of Day	3 - 5	15,414	8.2	+38%
Top 10 Decreases					
B	Age	18 - 24	10,590	0.0	-90%
A	Age	18 - 24	5,260	0.0	-90%
B	Location	Bangladesh	10,376	0.0	-90%
A	Location	South Africa	61,756	5.0	-90%
B	Location	Ethiopia	5,609	0.0	-90%
B	Location	Philippines	20,574	0.0	-90%
B	Location	Pakistan	14,007	0.0	-90%
B	Location	Nigeria	13,956	0.0	-90%
B	Hour of Day	18 - 20	7,426	0.0	-90%
A	Location	Mexico	10,040	1.0	-88%

Note: total of 53 non-zero bid adjustments.

Bid adjustments are only calculated where we have more than 5,000 clicks.